One Acre Fund SMS Repayment Trial

Analysis Insights Report

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Executive Summary

An SMS-based repayment nudge trial was conducted across 462,885 enrolled farmers, testing six message variants against a no-SMS control group. After excluding 190,727 farmers who had already qualified before the trial (pre-trial qualifiers whose behaviour cannot be influenced by messaging), the effective analytical population was 272,158 in-scope farmers. Results confirm that structured SMS messaging lifts loan repayment qualification rates by up to 2.07 percentage points over control, with five of the six treatment arms reaching statistical significance. The findings provide a clear, actionable roadmap for scaling high-performing message types and retiring low-value interventions.

Key Findings

1. SMS Messaging Demonstrably Increases Repayment Qualification

The control group (no SMS) recorded a 63.68% qualification rate among in-scope farmers. Five out of six SMS treatment arms exceeded this baseline with statistically significant lifts ($p < 0.05$ in all five cases). T2 Benefits delivered the highest primary-outcome lift at +2.07 percentage points, followed closely by T5 Social (+1.94pp) and T1 Basic (+1.84pp). These are not marginal improvements given the scale: at 181,448 treated in-scope farmers, even a 1.84pp lift translates to approximately 3,339 additional qualified farmers per campaign cycle.

Arm	Qual Rate	Lift (Dec31)	Dec15 Lift	Significant?	Verdict
Control (T0)	63.68%	Baseline	Baseline	Baseline	Baseline
T1 Basic	65.53%	+1.84pp	+1.82pp	Yes	Scale
T2 Benefits	65.76%	+2.07pp	+2.03pp	Yes	Scale
T3 Pay Early	65.50%	+1.82pp	+2.43pp	Yes	Scale
T4 Pay Little	64.92%	+1.24pp	+0.99pp	Yes	Consider
T5 Social	65.62%	+1.94pp	+2.35pp	Yes	Scale
T6 Group Leader	64.03%	+0.34pp	+0.15pp	No	Retire

2. T3 "Pay Early" Is the Strongest Driver of Early Payment

While T2 Benefits leads on the primary Dec 31 qualification metric, T3 Pay Early dominates early payment (Dec 15) with a +2.43pp lift over control. This is a strategically valuable distinction: early payment improves cash flow predictability for operations. T5 Social also performs strongly on Dec 15 (+2.35pp), suggesting that both urgency framing and social proof are effective levers when timed correctly.

3. T6 Group Leader Channel Should Be Retired

T6, which routes the repayment nudge through group leaders rather than sending SMS directly to farmers, produced a lift of only +0.34pp on Dec 31 and +0.15pp on Dec 15, neither of which approaches statistical significance ($p = 0.34$). This outcome likely reflects message dilution through an intermediary. Continuing to invest in this channel produces no measurable benefit and represents wasted messaging budget.

4. Targeting Segment: Who Responds Most to SMS

Subgroup analysis reveals three high-value targeting segments:

- New farmers (first enrolled season): Baseline qualification rate of ~58%, lowest of all experience cohorts. SMS lifts are proportionally largest in this group, making them the highest-priority target for direct messaging.
- Farmers who did not qualify in 2017: This cohort has both the lowest baseline and the most room to improve. Treatment effects in this group are consistently above the full-population average.
- Low-credit farmers (Q1 quartile): Farmers with the smallest credit amounts sit at a ~55% baseline, well below mid- and high-credit peers (~72-76%). Targeted messaging here yields the highest marginal qualification gains.
- Group leaders: Qualify at 77-79% regardless of any treatment arm. SMS adds negligible value for this segment. They should be deprioritised from the direct-message budget.

5. Projected Impact at Scale

Scaling T2 Benefits to all 181,448 treated in-scope farmers would generate approximately 3,756 additional qualified farmers per cycle. At an assumed operational value of NGN 500 per additional qualifier, this represents NGN 1.88M in incremental value per campaign. T5 Social and T1 Basic follow with approximately 3,519 and 3,338 additional qualifiers respectively. Combined deployment of the top three arms, intelligently targeted at new, first-season, and low-credit farmers, could conservatively push the aggregate lift above 5,000 additional qualifiers per cycle.

Recommendations

- Scale T2 Benefits, T5 Social, and T3 Pay Early as the primary SMS content strategy. These three arms consistently outperform across both outcome windows.
- Retire T6 Group Leader. No significant effect was detected; redirect that budget to direct farmer messaging.
- Prioritise new farmers (season 1), 2017 non-qualifiers, and Q1 low-credit farmers in targeting logic to maximise return per SMS sent.
- Integrate early-payment incentive framing from T3 into the main messaging calendar to improve Dec 15 outcomes alongside Dec 31 rates.
- Conduct a cost-per-qualified-farmer analysis using actual SMS unit costs to rank arms on ROI rather than lift alone, which will sharpen scaling decisions.